EXPERIMENT NO. 10

CONFIGURATION OF FIREWALL USING IP TABLES

AIM

To configure and manage firewall rules for securing a network.

OBJECTIVES

- To understand firewall and packet filtering concepts.
- To configure IP Tables for traffic control.
- To allow or deny traffic based on rules.
- To secure network services using firewall policies.
- To verify firewall operation through testing.

THEORY

A firewall monitors and controls incoming and outgoing traffic using predefined rules. It protects trusted networks from unauthorized traffic.

IPTables is a Linux command line utility used for packet filtering, NAT, traffic control and access control.

SOFTWARE / MATERIALS REQUIRED

- Ubuntu/Linux Operating System
- Root Privileges
- Terminal Access

IMPORTANT COMMANDS / PROGRAM STEPS

```
sudo iptables -L
sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
sudo iptables -A INPUT -p icmp -j DROP
sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
sudo iptables-save
sudo iptables -L -v
```

PROCEDURE

1. Check existing IPTables rules.
2. Allow required services such as SSH or HTTP.
3. Block unwanted traffic such as ICMP if required.
4. Save the firewall rules.
5. Verify rules using verbose listing.
6. Test allowed and blocked traffic.

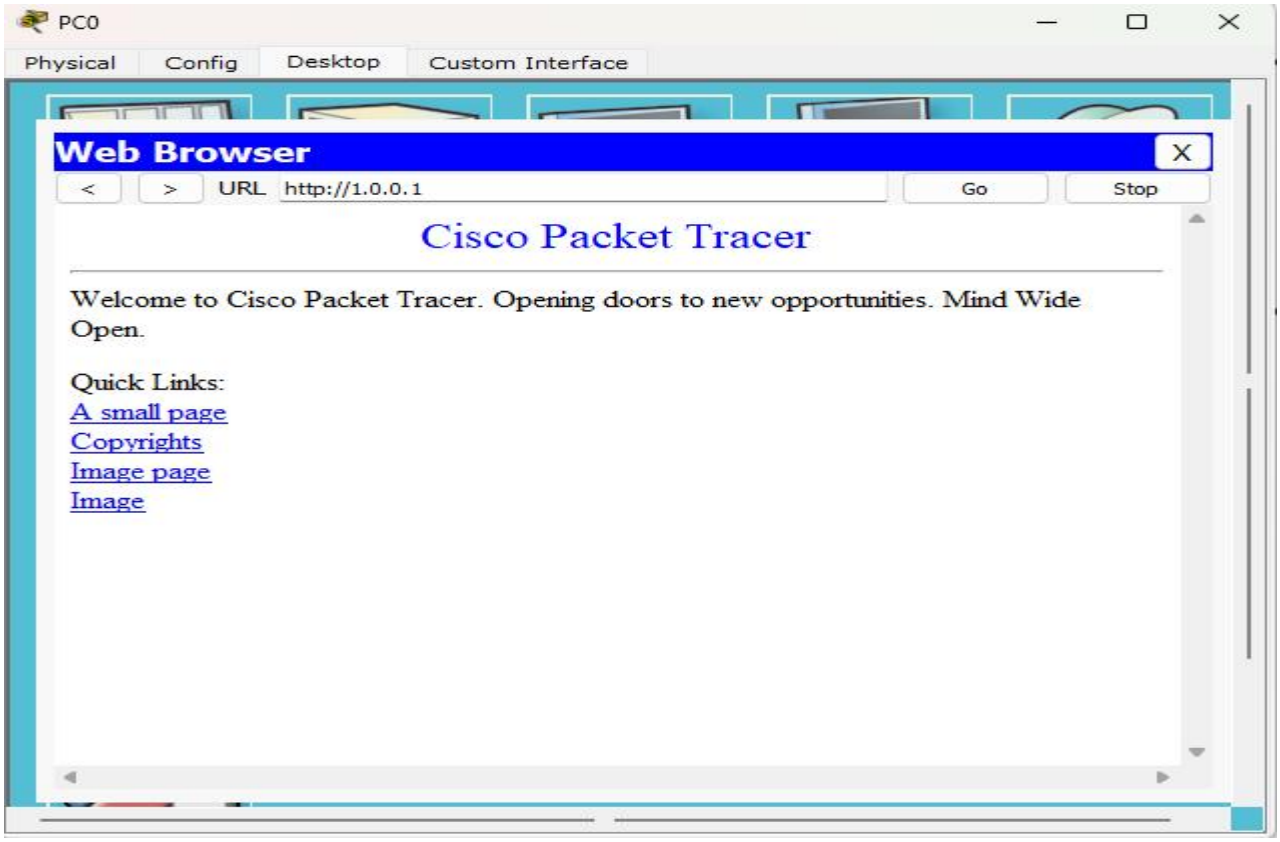
OBSERVATION / COMPARISON TABLES

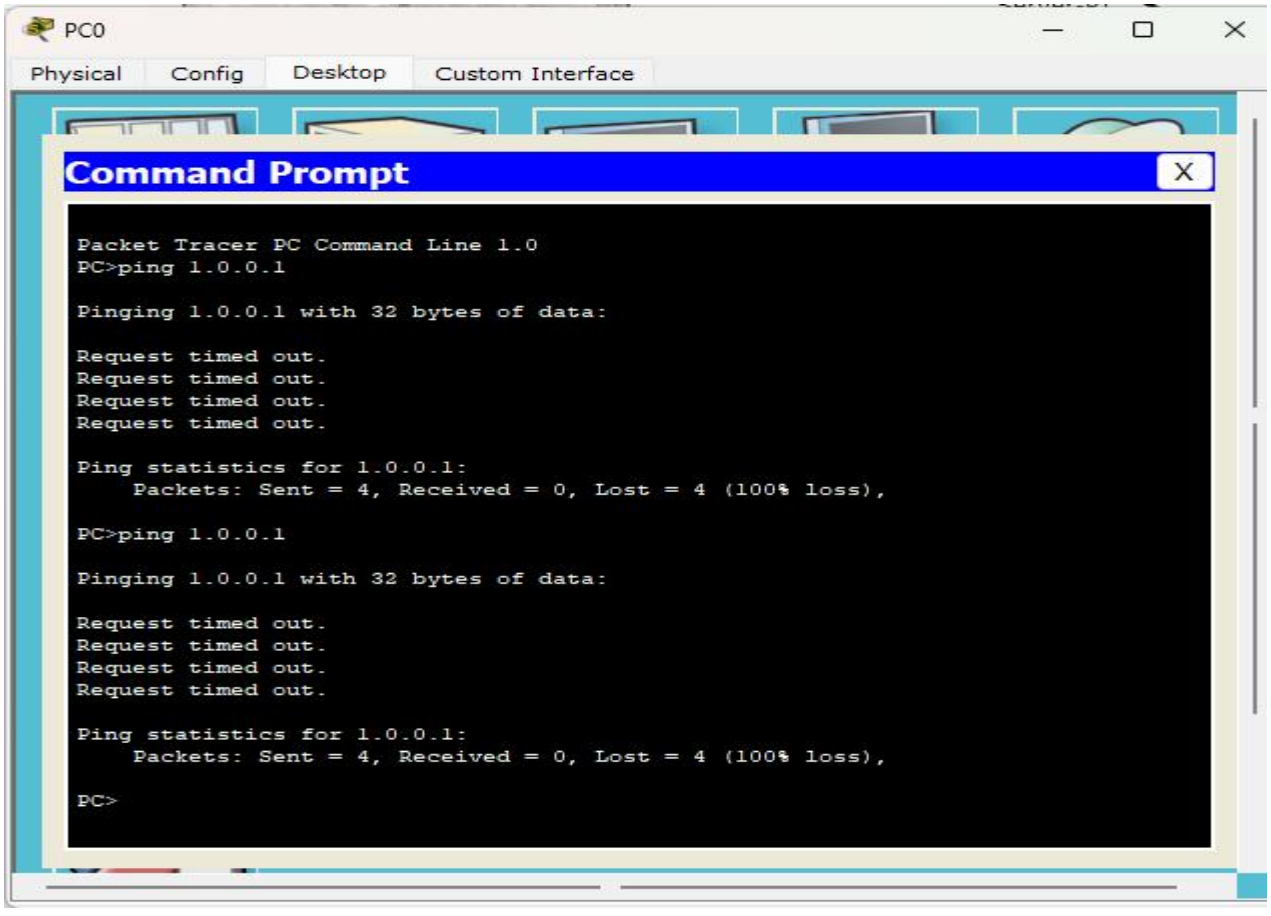
Chain	Function
INPUT	Controls incoming packets
OUTPUT	Controls outgoing packets
FORWARD	Controls forwarded packets

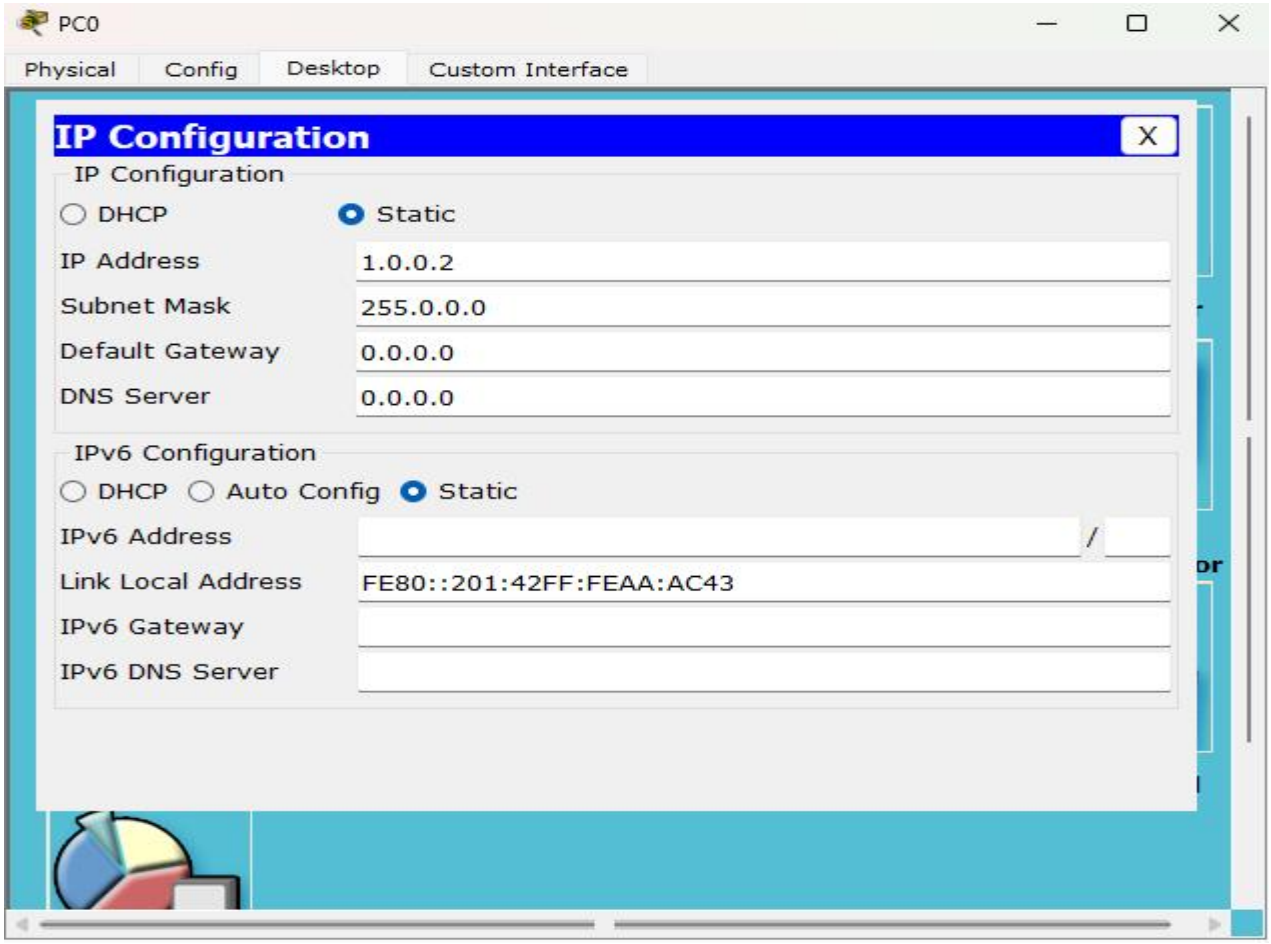
Action	Description
ACCEPT	Allows traffic
DROP	Silently drops traffic
REJECT	Rejects traffic with notification
LOG	Records packet information

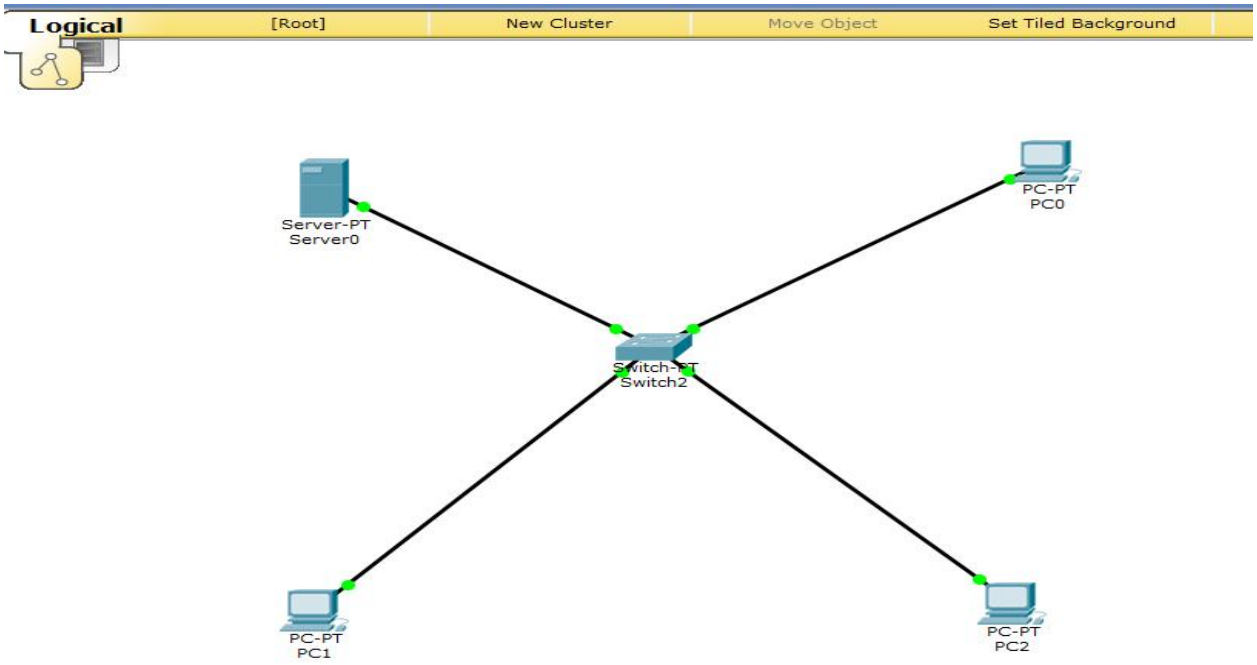
Rule Configured	Purpose	Status
Allow SSH	Remote Access	
Allow HTTP	Web Service	
Block ICMP	Disable Ping	
Default Policy	Security Control	

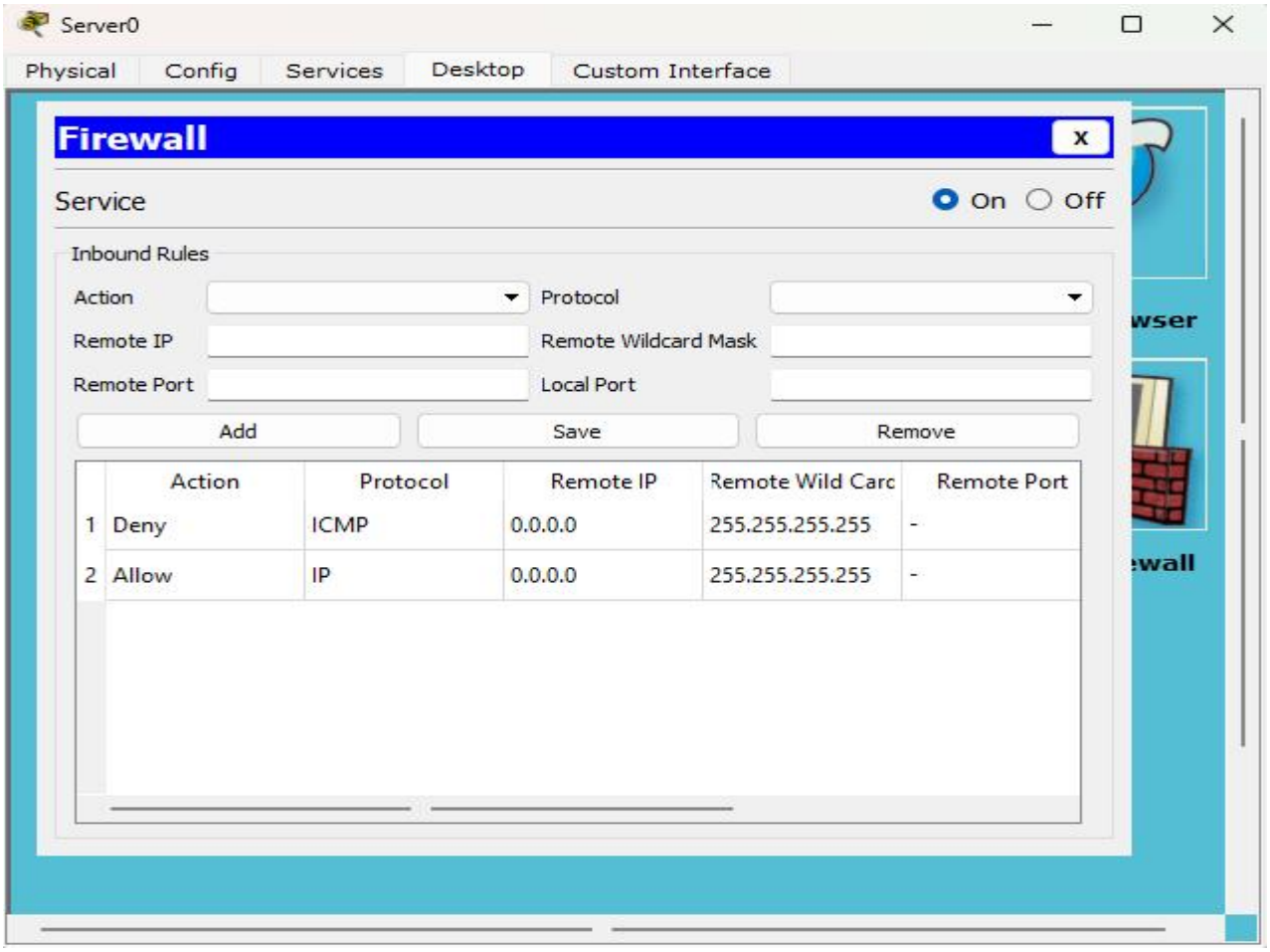
RESULT:

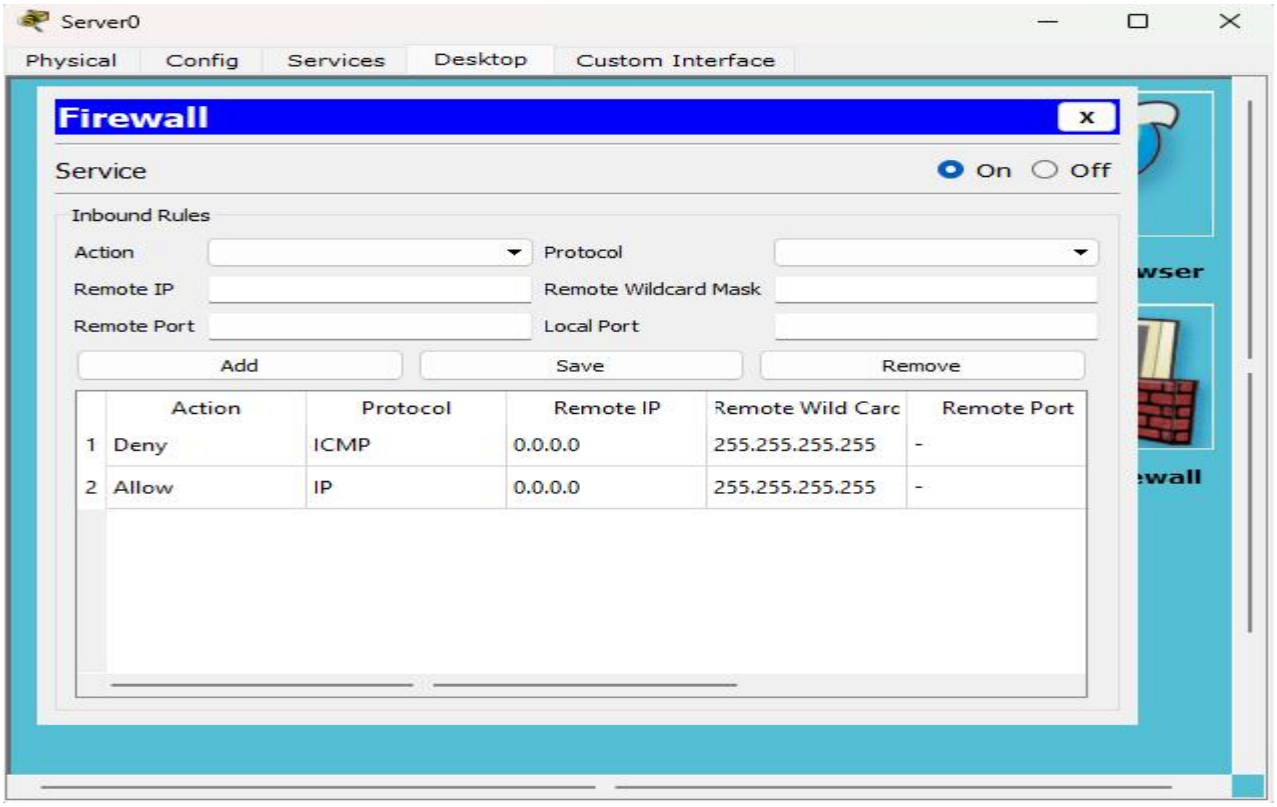

PC0
Physical
Config
Desktop
Custom Interface
Web Browser
URL
http://1.0.0.1
Go
Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image


PC0
Physical
Config
Desktop
Custom Interface
Command Prompt
Packet Tracer PC Command Line 1.0
PC>ping 1.0.0.1
Pinging 1.0.0.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 1.0.0.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PC>ping 1.0.0.1
Pinging 1.0.0.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 1.0.0.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PC>


PC0
Physical
Config
Desktop
Custom Interface
IP Configuration
X
IP Configuration
DHCP
Static
IP Address
1.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::201:42FF:FEAA:AC43
IPv6 Gateway
IPv6 DNS Server


Logical
[Root]
New Cluster
Move Object
Set Tiled Background
Server-PT
Server0
PC-PT
PC0
Switch-PT
Switch2
PC-PT
PC1
PC-PT
PC2


Server0
Physical
Config
Services
Desktop
Custom Interface
Firewall
X
Service
On
Off
Inbound Rules
Action
Protocol
Remote IP
Remote Wildcard Mask
Remote Port
Local Port
Add
Save
Remove
Action
Protocol
Remote IP
Remote Wild Carc
Remote Port
1 Deny ICMP 0.0.0.0 255.255.255.255 -
2 Allow IP 0.0.0.0 255.255.255.255 -


Server0
Physical
Config
Services
Desktop
Custom Interface
Firewall
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2 Allow IP 0.0.0.0 255.255.255.255 -

CONCLUSION:

The experiment was successfully performed to configure and manage a firewall using **IPTables**. The required network services were allowed, unwanted traffic was blocked, and the configured rules were verified using IPTables commands. Thus, IPTables provides an effective method for controlling network traffic and improving the security of a Linux system.